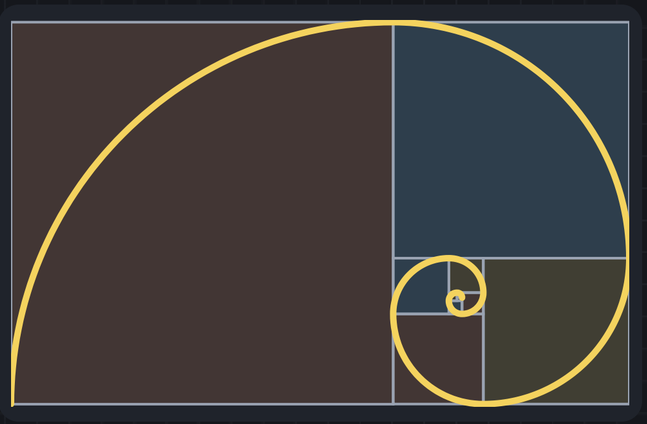
Pi & Planck

FAMOUS CONSTANT

The Golden Ratio

One number, φ = 1.618..., links a pentagon, Fibonacci numbers and sunflower seeds. Some famous claims about it are myths.

THE GOLDEN RECTANGLE



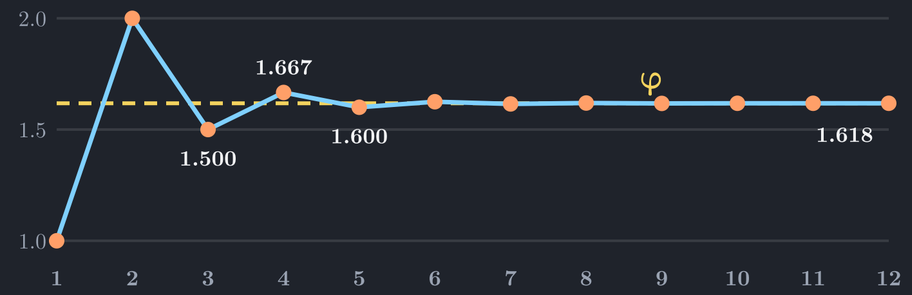
Cut off a square and what is left has the same shape. Repeat forever.

$$\varphi = (1 + \sqrt{5}) \,/\, 2 = 1.6180339887\ldots$$

$$\varphi^2 = \varphi + 1$$

WHERE IT REALLY SHOWS UP

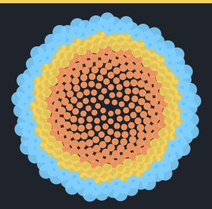
Fibonacci ratios F(n+1) / F(n) close in on φ





Pentagon

diagonal ÷ side = φ



Sunflower seeds

turn 137.5° per seed

HISTORY AND MYTH

c. 300 BC: Euclid

The Elements defines a line cut in “extreme and mean ratio”, the first known description of what we call the golden ratio.

Most irrational number

φ = 1 + 1/(1 + 1/(1 + ...)): the continued fraction is all ones, so no fraction approximates it quickly.

Myth check: the nautilus

Measured Nautilus shells grow by 1.24 to 1.43 per quarter turn, mean 1.33. A golden spiral grows by 1.618.

1.618 the ratio of the longer part to the shorter: a/b = (a + b)/a

137.5° the golden angle, 360° ÷ φ²: the turn per seed in a sunflower

1.33 mean growth per quarter turn measured on Nautilus shells

Sources: Euclid, Elements VI, Def. 3 · C. Falbo, The Golden Ratio: A Contrary Viewpoint, College Mathematics Journal 36 (2005) 123–134 · H. Vogel, Mathematical Biosciences 44 (1979) 179–189 · Numbers and diagrams computed for this graphic